

Line-by-Line Explanation: Construct Binary Tree from Preorder and Inorder Traversal

1. Preorder traversal gives the root first.
2. Inorder traversal divides left and right subtrees.
3. HashMap stores inorder values for fast lookup.
4. preorderIndex tracks the current root.
5. Base case stops recursion when range is invalid.
6. Root is created using preorder value.
7. Inorder index splits left and right subtree.
8. Recursive calls build left and right subtrees.
9. Final root is returned forming the tree.

Time Complexity: $O(n)$ Space Complexity: $O(n)$